

Data Link Protocols

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Data
Communication
Week 7

- Protocol is a set of rules used to perform the necessary operations during data flow.



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Data Link Controls (DTC)

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- Components
 - Software
 - Hardware (UART/USART)
- Tasks
 - Synchronous Transmission
 - Physical Level: Common timing signal
 - Logical Level: Special bit or bit-arrays
 - Flow Control
 - Error Control
 - Transmission Rules
 - Which one is sender/receiver?
 - Whether frames are for data or control purposes



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Asynchronous Protocols

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- Used for connections made with a **modem**
- Logical Sync.
 - Start bit
 - End bit
- There may be gaps of variable sizes between data blocks.
- Cons: Additional data
- Cons: Slow transmission
- Most Common Examples:
 - X-Modem
 - Y-Modem
 - Z-Modem
 - Kermit



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Synchronous Protocols

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- Instead of start and end bits, FLAGS is used
 - Synchronization premise (SYN) bit-arrays
 - Synchronization successor (EOT) bit-arrays
- 2 types:
 - Physical Level Synchronization
 - Common Clock Signal
 - Logical Level Synchronization (in Data Link Layer)
 - FLAGS
- Thanks to the fast transmission
 - Used in LAN, MAN and WAN technologies
- The protocol perceives the data sent as a consecutive byte sequence.
 - According to the coding system used (ASCII or EBCDIC)



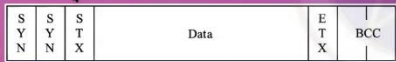
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Synchronous Protocols - Byte Oriented

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- *Character Oriented
- ASCII or EBCDIC coding
- 2 types of frames
 - Control frames
 - Data frames

Control Chars	Explanation
SYN	Synchronous idle (makes the channel active)
PAD	Frame PAD (used for completion)
DLE	Data Link Escape (Escape char for control chars)
ENQ	Enquiry (Request)
SOH	Start of Heading
STX	Start of Text
ITB	End of Intermediate Block
ETB	End of Transmission Block
ETX	End of Text
EOT	End of Transmission
BCC	Block Check Count (LRC -> 1 byte, CRC -> 2 bytes)
ACK0	Acknowledge Even Numbered Blocks
ACK1	Acknowledge Odd Numbered Blocks
WACK	Wait Before Transmitting
TTD	Temporary Text Delay (While filling buffer in sender side)
RVI	Reverse Interrupt (Request for urgent response)
NUL	(Filling spaces)



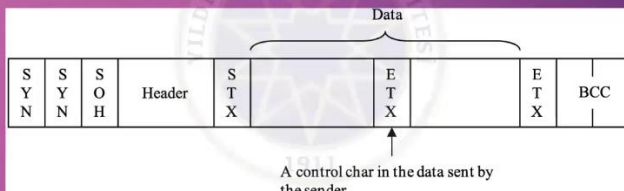
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Synchronous Protocols - Byte Oriented - Data Frames

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Header includes address



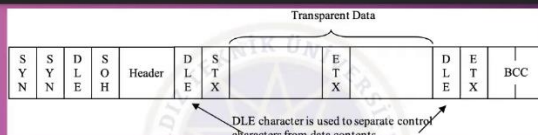
A control char in the data sent by the sender

Data Transparency ? (in voice and image data)

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Synchronous Protocols - Byte Oriented - Data Frames

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DLE character is used to separate control characters from data contents

DLE before STX, ETX, ETB, ITB, SOH control chars

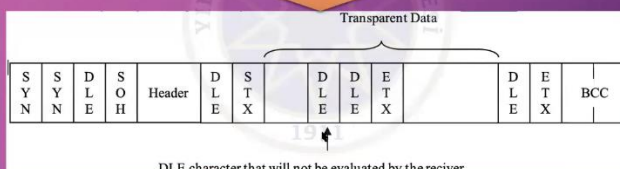
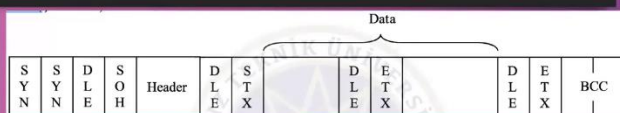


If data includes DLE char?

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Synchronous Protocols - Byte Oriented - Data Frames

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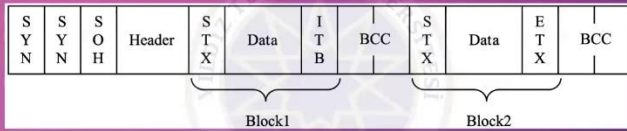


DLE character that will not be evaluated by the receiver

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Synchronous Protocols - Byte Oriented - Multi Block Frames

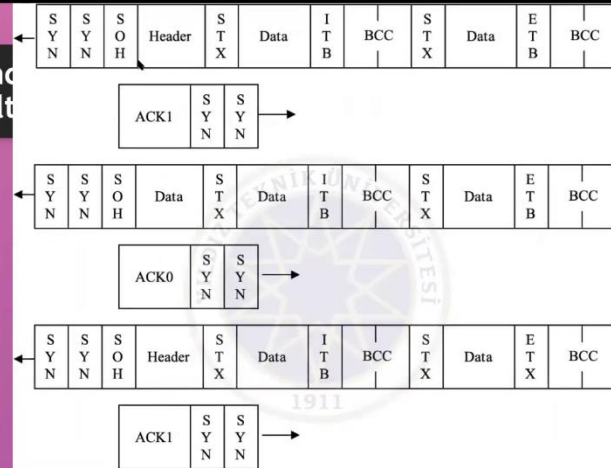
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Synchronous Protocols - Byte Oriented - Multi Block Frames

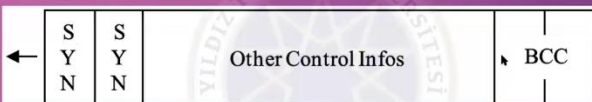
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Synchronous Protocols - Byte Oriented - Control Frames

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Synchronous Protocols - Byte Oriented - BSC (Binary Synchronous Communication)

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- IBM
- Used until the late 1960s
- General Specs:
 - Suitable for point-to-point or multi-point connections
 - Half duplex
 - Use STOP & WAIT for Flow Control and ARQ
 - Code Dependent

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Synchronous Protocols - Bit Oriented

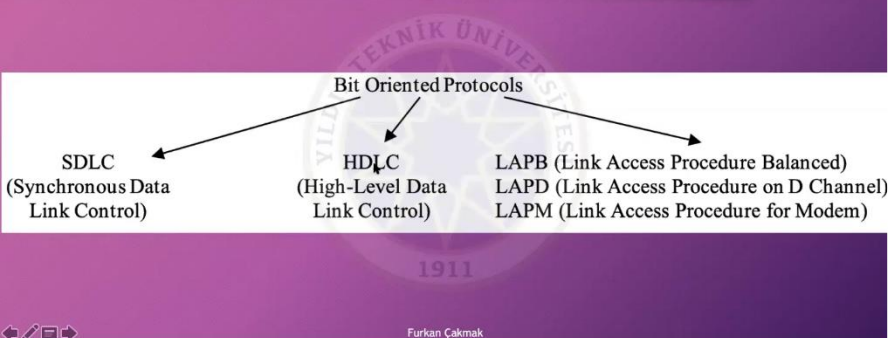
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- Used more actively than byte oriented protocols
 - Fit more information into a shorter frame size
 - Less faced with data transparency problem
- All bit oriented protocols reference HDLC (High Level Data Link Control)
 - ISO
- General Specs:
 - Suitable for point-to-point or multi-point connection.
 - Support Half duplex and Duplex
 - Use Sliding Windows for Flow Control and ARQ
 - Code Independent

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Synchronous Protocols - Bit Oriented

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Synchronous Protocols - Bit Oriented - HDLC (High Level Data Link Control)

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- Concepts used in HDLC
 - Station types
 - Primary
 - Secondary
 - Combined
 - Configuration
 - Unbalanced
 - Balanced (not defined in HDLC)
 - Modes of communication
 - NRM (Normal Response Mode): Unbalanced. The usual primary/secondary relationship
 - ARM (Asynch. Response Mode): Unbalanced. Secondary station transmit data without primary station's permission if line is available.
 - ABM (Asynch. Balanced Mode): Balanced

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Synchronous Protocols - Bit Oriented - HDLC (High Level Data Link Control)

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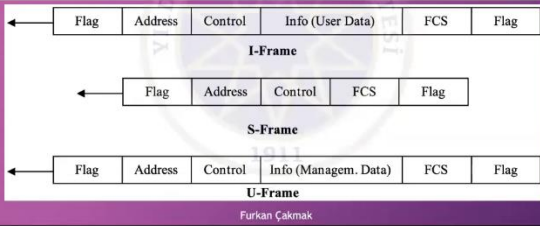
Modes of Communication	NRM	ARM	ABM
Station Types	Primary/Secondary	Primary/Secondary	Combined
Transmission Starter	Primary	Primary/Secondary	Any of them

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Synchronous Protocols - Bit Oriented - HDLC Frame Structure

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I-Frame	Information	It is used to carry user data and related control information.
S-Frame	Supervisory	It is the type of frame used at the data link layer to perform functions such as error and flow control.
U-Frame	Unnumbered	It is a special purpose management frame used to provide system management.



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Synchronous Protocols - Bit Oriented - HDLC Frame Structure - FLAG Field

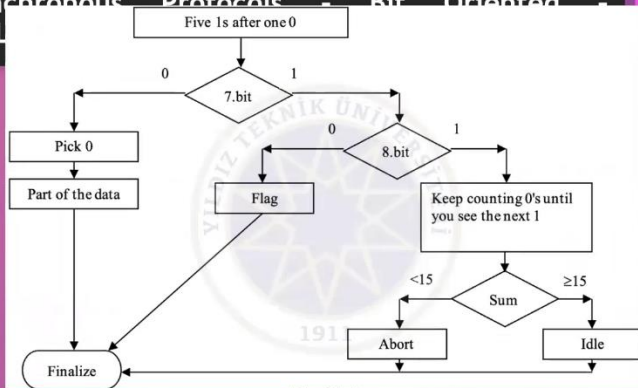
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- Consist of 8-bits
 - 01111110
- Determinin start and end point of frames
- Ensure synchronicity
- Critical point for data transparency
 - Bit stuffing
 - Sender add a 0-bit between 1's (like escape char)

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Synchronous Protocols - Bit Oriented - HDLC Frame Structure - FLAG Field

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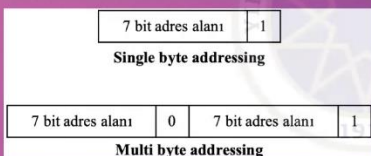


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Synchronous Protocols - Bit Oriented - HDLC Frame Structure - ADDRESS Field

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- Contains the address of the secondaries.
- Address data is in Network Layer (Third Layer)
 - Used to determine whether;
 - Is it command? or Is it answer?

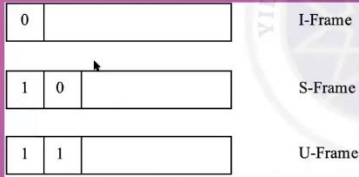


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Synchronous Protocols - Bit Oriented - HDLC Frame Structure - CONTROL Field

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- 1 or 2 byte length
 - If $w=7$ in sliding window technique, length is 1
 - If $w=127$ in sliding window technique, length is 2
 - In applications with high line delays such as WAN

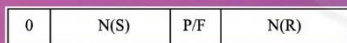


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Synchronous Protocols - Bit Oriented - HDLC Frame Structure - I-Frame

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- If $w=7$;
 - Each I-Frame contains 2 three-bit fields for flow and error control.
 - N(S): Sequence number of the **sending** window
 - Like ACK
 - N(R): Sequence number of the **receiving** window
 - If last frame is error-free, N(R) contains next frame's number
 - If last frame has errors, N(R) contains number of faulty frame
- If $w=127$
 - Each I-Frame contains 2 seven-bit fields for flow and error control.

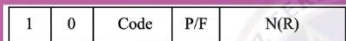


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Synchronous Protocols - Bit Oriented - HDLC Frame Structure - S-Frame

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- S-Frame means that neither side has data to send to the other.



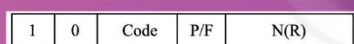
Code	Abbreviation	Explanation
00	RR	Receive Ready
01	REJ	Reject -- (go back n)
10	RNR	Receive Not Ready
11	SREJ	Selective Reject -- (selective reject)

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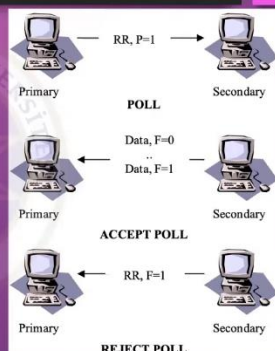
Synchronous Protocols - Bit Oriented - HDLC Frame Structure - S-Frame - RR

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- It has four basic uses:
 - ACK
 - POLL
 - $P=1$
 - REJECT POLL
 - $F=1$
 - ACCEPT SELECT
 - $F=1$



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Synchronous Protocols - Bit Oriented - HDLC Frame Structure - S-Frame - RNR

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- It has three basic uses:

- ACK
- SELECT
 - P=1
 - If primary sends RR and P=1, it means POLL.
- REJECT SELECT
 - F=1

1	0	Code	P/F	N(R)
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Synchronous Protocols - Bit Oriented - HDLC Frame Structure - S-Frame - REJ, SREJ

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- Used to return negative feedback from the receiver

- REJ: Go Back N ARQ
 - It is used to inform the sender that the frame whose number is written in the N (R) field and the frames that come after it did not reach the receiver or that it received incorrectly, and to ensure that it is sent again.
- SREJ: Selective Reject ARQ
 - It is used in the N (R) field to inform the sender that the data frame whose number is written on it did not reach the receiver or that it was received incorrectly and to send it again.

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Synchronous Protocols - Bit Oriented - HDLC Frame Structure - U-Frame

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- Used to provide session control
- The data area is used to carry information required for system management functions
- 1st code field consist of 2 bits
- 2nd code field consist of 3 bits
- $2^5 = 32$ different state
- These states (commands and answers) can be collected in 5 different categories:
 - Mode setting
 - Unnumbered exchange
 - Disconnection
 - Initialization
 - Miscellaneous

1	1	Code	P/F	Code
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Synchronous Protocols - Bit Oriented - HDLC Frame Structure - U-Frame - Mode Setting

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- How will the transmission be?
 - 00-001 (SNRM-Set Normal Response Mode) -> w=7
 - 11-000 (SARM-Set Async. Response Mode) -> w=7
 - 11-100 (SABM-Set Async. Balanced Mode) -> w=7
 - 11-011 (SNRM Extended) -> w=127
 - 11-010 (SARM Extended) -> w=127
 - 11-110 (SABM Extended) -> w=127

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Synchronous Protocols - Bit Oriented - HDLC Frame Structure - **U-Frame** - **Unnumbered Exchange**

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- Data connection information exchange
 - 00-100 (UP-Unnumbered Poll); POLL request
 - 00-000 (UI-Unnumbered Info); exchange of date/time information to be used for sync.
 - UI would be a command or answer.
 - If it is used for command, it transports list of parameters to be used for transmission.
 - If it is used for answer, it carries information that determines the capability of the receiver.
 - 00-110 (UA-Unnumbered Ack); Sent in response to the UP command

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Synchronous Protocols - Bit Oriented - HDLC Frame Structure - **U-Frame** - **Disconnection**

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- There are 3 types of disconnection command.
 - 00-010 (DISC): **Sent by the first side** to terminate the connection to the other.
 - 00-010 (RD): It is used to notify the request to terminate the connection **from the second station** to the first.
 - 11-000 (DM): **When the address is sent** from the specified station to the station wishing to establish the connection, it is sent as **negative feedback** information to the **mode setting command**.

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Synchronous Protocols - Bit Oriented - HDLC Frame Structure - **U-Frame** - **Initialization**

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- Used for initializations for all sides:
 - 10-000 with P (SIM-Set Initialization Mode)
 - Command is sent from the first station to the second
 - UI command will be sent in response to SIM command
 - 10-000 with F (RIM-Request Initialization Mode)
 - It means that «I am waiting SIM command»
 - It is used when the second station cannot respond to the mode setting command without receiving the SIM command from the first.

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Synchronous Protocols - Bit Oriented - HDLC Frame Structure - U-Frame - Miscellaneous

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- 11-001 (RESET)
 - This is usually sent in response to a received FRMR code.
 - Explains that the secondary station must do the same
- 11-101 (XID)
 - Emphasizes that a **self-determining information is requested** from the secondary station.
 - Like questioning what your address is
- 10-001 (FRMR)
 - Used to determine that a **syntax error** was encountered in the received frame.

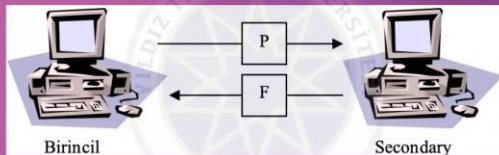
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Synchronous Protocols - Bit Oriented - HDLC Frame Structure - S-Frame - P/F (Poll/ Final)

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- P/F bit is always 1



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